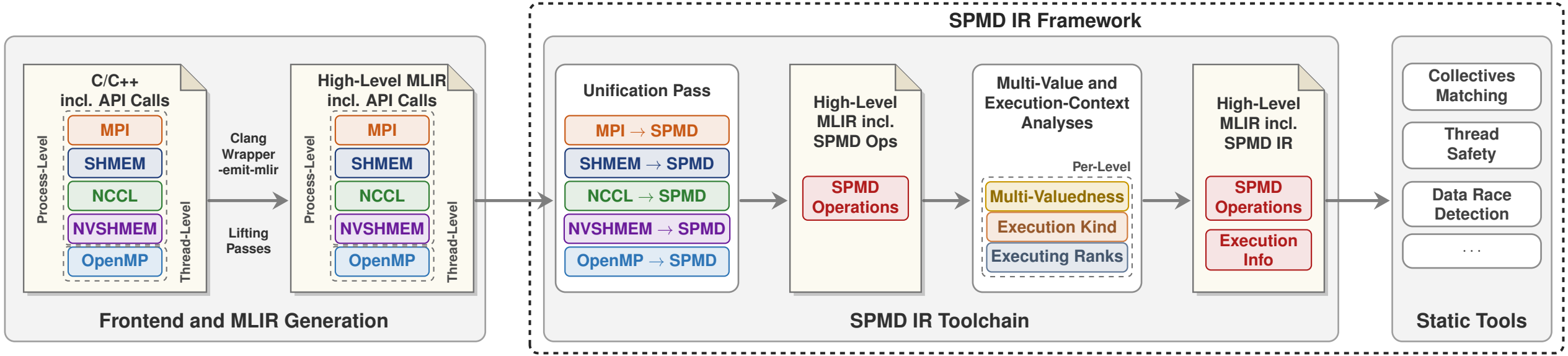
SPMD IR Framework
C/C++
incl. API Calls
Process-Level
MPI
SHMEM
NCCL
NVSHMEM
OpenMP
Thread-Level
Clang
Wrapper
-emit-mlir
Lifting
Passes
High-Level MLIR
incl. API Calls
Process-Level
MPI
SHMEM
NCCL
NVSHMEM
OpenMP
Thread-Level
Frontend and MLIR Generation
Unification Pass
MPI → SPMD
SHMEM → SPMD
NCCL → SPMD
NVSHMEM → SPMD
OpenMP → SPMD
High-Level
MLIR incl.
SPMD Ops
SPMD
Operations
Multi-Value and
Execution-Context
Analyses
Per-Level
Multi-Valuedness
Execution Kind
Executing Ranks
High-Level
MLIR incl.
SPMD IR
SPMD
Operations
Execution
Info
SPMD IR Toolchain
Collectives
Matching
Thread
Safety
Data Race
Detection
...
Static Tools